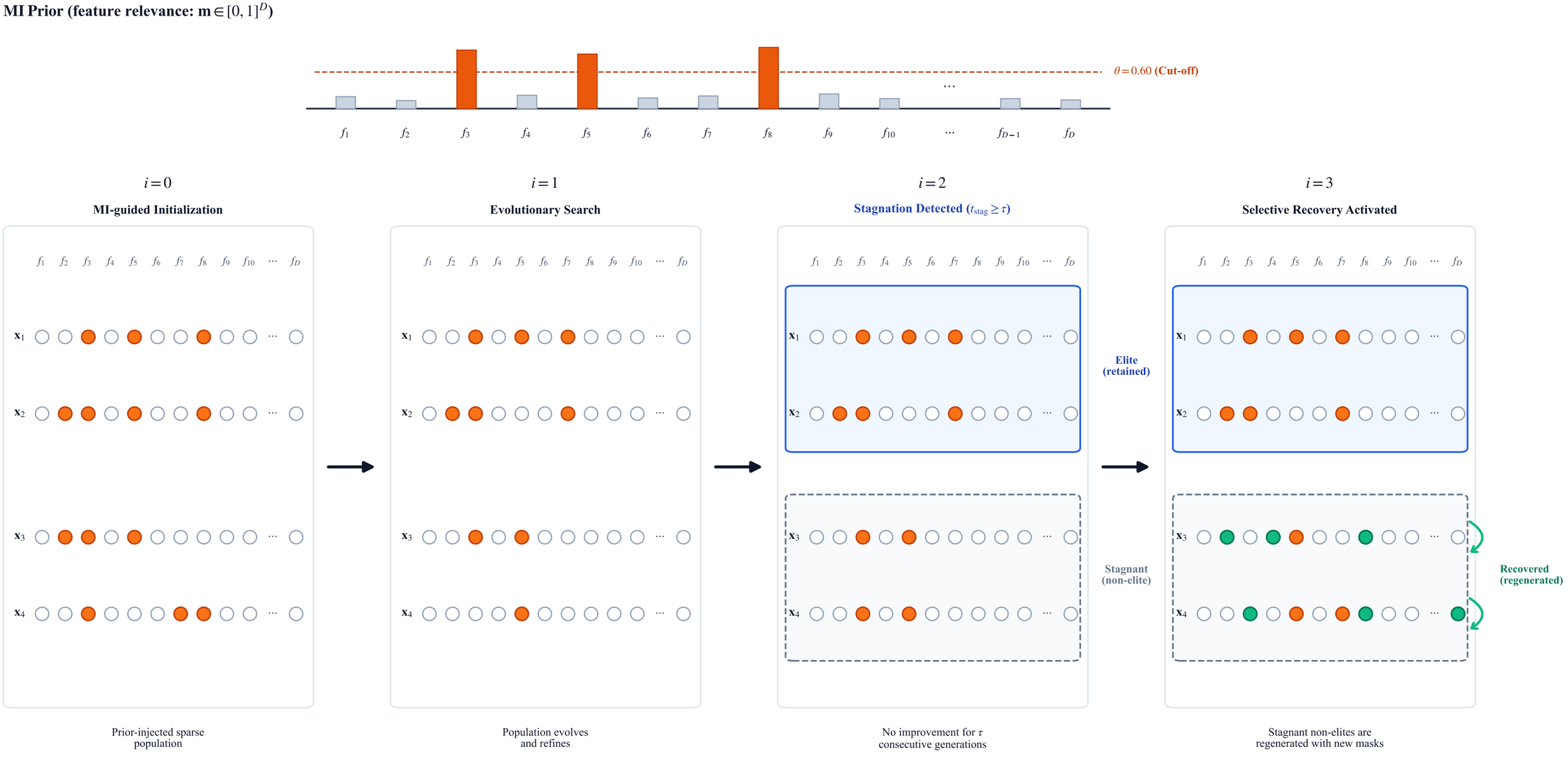

MI Prior (feature relevance: $\mathbf{m} \in [0,1]^D$)
$\theta = 0.60$ (Cut-off)
f_1 f_2 f_3 f_4 f_5 f_6 f_7 f_8 f_9 f_{10} ... f_{D-1} f_D
$i = 0$
MI-guided Initialization
$i = 1$
Evolutionary Search
$i = 2$
Stagnation Detected ($t_{\text{stag}} \geq \tau$)
$i = 3$
Selective Recovery Activated
$\mathbf{x}_1$ $\mathbf{x}_2$ $\mathbf{x}_3$ $\mathbf{x}_4$
Elite (retained)
Stagnant (non-elite)
Recovered (regenerated)
Prior-injected sparse population
Population evolves and refines
No improvement for τ consecutive generations
Stagnant non-elites are regenerated with new masks
1 (selected)
0 (unselected)
Recovered feature ($\mathbf{m} + \varepsilon_i$)